



NARAYANA
COACHING CENTER
IIT- JEE/NEET/FOUNDATION

45
YEARS
OF EXCELLENCE

Sec: XI-STU-IC-IIT

Time : 3 Hrs.

FTM – 2

Date :17-06-2024

Max.Marks:300

JEE MAIN – 2022Model

PHYSICS:

Section	Question Type	+Ve Marks	–Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 1 – 20)	Questions with Single Correct Choice	4	–1	20	80
Sec – II (Q.N : 21 – 30)	Questions with Numerical Answer Type(+ / – Decimal Number)	4	0	10	20
Total				30	100

CHEMISTRY :

Section	Question Type	+Ve Marks	–Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 31 – 50)	Questions with Single Correct Choice	4	–1	20	80
Sec – II (Q.N : 51 – 60)	Questions with Numerical Answer Type(+ / – Decimal Number)	4	0	10	20
Total				30	100

MATHEMATICS:

Section	Question Type	+Ve Marks	–Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 61 – 80)	Questions with Single Correct Choice	4	–1	20	80
Sec – II (Q.N : 81 – 90)	Questions with Numerical Answer Type(+ / – Decimal Number)	4	0	10	20
Total				30	100

Subject	Exam Syllabus
Physics	40% : KINEMATICS: Frame of Reference, Mathematical Expression and definition of position, distance, and displacement Avg velocity & instantaneous velocity, Avg speed & instantaneous speed, acceleration 40% : Unit and Dimension: Significant figures, Errors in measurements Problems discussion from material. 20% : MATHEMATICAL PHYSICS: Application of differentiation in physics, Maxima & Minima Integration and its application in physics. Unit and Dimension: Introduction of physics, physical quantities, and physical laws Fundamental and derived quantities and ways to write dimension, Unit and Dimension Uses of dimensional analysis (Principle of homogeneity).
Chemistry	40% : Question practice on concentration Concept of limiting reagent: Use of stoichiometry in balanced equation and limiting reagent concept. 40% : Concentration terms: Molarity, molality, mole fraction and their interconversions, relation between density and molarity. Concentration terms and their interconversions, relation between density and molarity 20% : Nature of Matter, Properties of matter and their measurement SI unit, Mass weight, Volume temperature, need for standard reference, Scientific notation, Precision and accuracy, Significant figure and calculation involving significant figure. Law of chemical combination: (i) Law of conservation of mass, (ii) Law of constant combination, (iii) Law of multiple proportion, (v) Gay Lussac's law, (vi) Avagadro law, (vii) Dalton's atomic theory atomic mass Mole concept: Calculation based on mole concept, concept of gram atom, gram molecule Percentage by mass (w/w), percentage by volume (v/v).
Maths	40%: TRIGONOMETRIC FUNCTIONS Angle, system of measurement of angles, trigonometric function, Values of trigonometric function for some specific angles Trigonometric ratios of allied angles, transforming product into sum or difference, Transforming the sum or difference into product. 40%: Introduction to HP, HM, Relation between AM, GM, and HM Difference method, Vn method, special series 20%: SEQUENCE AND SERIES Introduction of Sequence G.P., sumies, nth term of AP. Properties of A.P., Sum of n term of an AP, Arithmetic Mean nth term of G.P., sum of n term of GP. Properties of G.P., Geometric mean, A.G.P.

Space for Rough Work

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

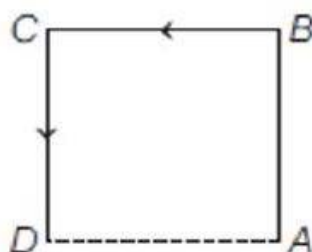
This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The side of a cube is measured by a Vernier calliper, whose 10 divisions of the vernier scale are equal to 8 divisions of main scale. If the main scale reads 10 mm and second division of vernier scale coincides with one of the main scale divisions, then the volume of the cube in appropriate significant figures is (take 1MSD = 1 mm)

A) 2.76 cm^3 B) 1.03 cm^3 C) 1.12 cm^3 D) 1.00 cm^3

2. A particle moves along the sides AB, BC, CD of a square of side 25 m with a velocity of 15 m/s. Its average velocity is



A) 15 m/s B) 10 m/s C) 7.5 m/s D) 5 m/s

3. The velocity of a body depends on time t according to the equation $v = 10 + 2t^2$. The acceleration of the body is

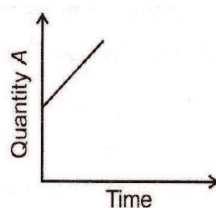
A) Independent of time t
B) Directly proportional to time t
C) Inversely proportional to time t
D) Directly proportional to square of time t

Space for Rough Work

4. Two trains, each of length 100 m are travelling in opposite direction with speed 15 m/s and 25 m/s. The time taken for crossing **each other** is

A) 4 s B) 2.5 s C) 5 s D) 2 s

5. If the variation of a physical quantity A with time is as shown in figures, then



- A) Quantity A is position if motion is uniform
 B) Quantity A is velocity if motion is uniform
 C) Quantity A is acceleration if motion is uniform
 D) Quantity A is position if motion is uniformly accelerated
6. For the one-dimensional motion of a particle, described by position $x = t + \cos t$ (where t is time in s and x is in m), the velocity of particle is zero at time t equals to
- A) 0.3 s B) 1 s C) 0.2 s D) 1.57 s
7. A car travels from A and B at a speed of 20 km h^{-1} and returns at a speed of 30 km h^{-1} . the average speed of the car for the whole journey is
- A) 5 km h^{-1} B) 24 km h^{-1} C) 25 km h^{-1} D) 50 km h^{-1}
8. The equation of state for n moles of a real gas is given by $\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = RT$. The dimension of PV is same as that of (where P, V and T are pressure, volume and temperature respectively)
- A) Force B) Surface tension C) Pressure D) Work
9. A body has speed $v, 2v$ and $3v$ in first $(1/3)$ rd of distance s , second $(1/3)$ rd of s and third $(1/3)$ rd of s respectively. Its average speed will be
- A) v B) $2v$ C) $\frac{18}{11}v$ D) $\frac{11}{18}v$

Space for Rough Work

10. Force applied by water stream depends on density of water (ρ), velocity of the stream (v) and cross-sectional area of the stream (A). The expression of the force should be
 A) ρAv B) ρAv^2 C) $\rho^2 Av$ D) $\rho A^2 v$
11. The velocity v of a moving particle varies with displacement as $x = \sqrt{v + 1}$, the acceleration of the particle at $x = 5$ unit will be
 A) $\sqrt{6}$ unit B) 24 unit C) 240 unit D) 25 unit
12. A body starts from the origin and moves along the X -axis such that the velocity at any instant is given by $(4t^3 - 2t)$, where t is in second and velocity in m/s. What is acceleration of the particle, when it is 2 m from the origin.
 A) 28 m/s² B) 12 m/s² C) 22 m/s² D) 10 m/s²
13. A particle is moving in a straight line such that its velocity is given by $v = 6 + 3t^2$. If at time $t = 0$, the particle is at origin, then average velocity during $t = 0$ to $t = 2$ s is (where v is in m/s and t is in s)
 A) 8 m/s B) 10 m/s C) -7 m/s D) -10 m/s
14. A drunkard is walking along a straight road. He takes 6 steps forward and 4 steps backward and so on. Each step is $\frac{1}{2}$ m long and takes 1 s. There is a pit on the road 4 m away from the starting point. The drunkard will fall into the pit after
 A) 10 s B) 16 s C) 6 s D) 8 s
15. Numerical value of Young's modulus of iron in S.I. unit is a and in CGS unit is b , then $\frac{a}{b}$ is
 A) 10 B) 10^{-1} C) 10^2 D) 10^{-2}
16. The measured value of mass and volume of a cubical body are (23.35 ± 0.01) g and (2.30 ± 0.01) cm³ respectively. The maximum percentage error in its density is nearly
 A) 0.02% B) 1% C) 2.7% D) 0.5%
17. The differentiation of $y = \sqrt{x} - \frac{1}{\sqrt{x}}$ w.r.t. x is
 A) $\frac{x+1}{2x^{3/2}}$ B) $\frac{1}{2}\sqrt{x}$ C) $2\sqrt{x}$ D) $\frac{2}{x^{3/2}}$

Space for Rough Work

18. A famous relation in physics relates the moving mass m to the rest mass m_0 of a particle in terms of its speed v and the speed of light c . A student recalls the relation almost correctly but forgets where to put the v . If he writes $m = \frac{m_0}{\left(1 - \frac{1}{c^2}\right)^{1/2}}$, then correct relation would be
- A) $m = \frac{m_0}{\left(1 - \frac{v}{c^2}\right)^{1/2}}$ B) $m = \frac{m_0}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$ C) $m = \frac{m_0}{\left(1 - \frac{1}{vc^2}\right)}$ D) $m = \frac{m_0}{\left(1 - \frac{1}{c^2v^2}\right)^{1/2}}$
19. If $Z = \frac{A}{B}$, then maximum absolute error in Z will be (ΔA and ΔB are absolute error in A and B respectively)
- (A) $A\Delta B + B\Delta A$ (B) $\Delta A + \Delta B$ (C) $\frac{\Delta A}{A} + \frac{\Delta B}{B}$ (D) $\frac{\Delta A}{B} + \frac{A\Delta B}{B^2}$
20. If the value of two temperatures are $T_1 = (40.3 \pm 0.1)^\circ\text{C}$ and $T_2 = (70.2 \pm 0.2)^\circ\text{C}$, then the value of $(2T_1 + 2T_2)$ upto proper error limit is
- (A) $(221.0 \pm 0.1)^\circ\text{C}$
 (B) $(221.0 \pm 0.6)^\circ\text{C}$
 (C) $(221.0 \pm 0.2)^\circ\text{C}$
 (D) $(221.0 \pm 0.3)^\circ\text{C}$

SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

21. If x stands for distance, then the value of n for which the given equation $\int \frac{dx}{\sqrt{a^2 - x^n}} = \sin^{-1} \frac{x}{a}$ is dimensionally correct, is_____.
22. In a new system of units, the speed of light in vacuum is 90000 and acceleration due to gravity is 1 **if the** numerical value of distance between the Sun and Earth in terms of the new unit **is $x/2$** , if light takes 8 min and 20 s to cover this distance. **Find the value of x** _____.
23. In an experiment, a physical quantity Z given by $Z = \frac{P^{1/2}Q^{1/3}}{R}$, is calculated with percentage errors in measurements of P , Q and R as 2%, 3% and 1% respectively. The maximum percentage error in Z will be_____.

Space for Rough Work

24. The rate of change in area (cm^2/sec) of a square of side 4 cm when its side is increasing at the rate of 0.01 cm per second is $\frac{x}{100}$. Find the value of x _____.
25. The radius of an air bubble is increasing at the rate of $\frac{1}{2}\text{cms}^{-1}$. the rate of increase in its volume when the radius is 1 cm is $m\pi \text{ cm}^2/\text{sec}$. Find the value of m _____.
26. If unit of mass becomes 2 times, the unit of length becomes 4 times and the unit of time becomes 4 times, then unit of Planck's constant becomes n times. Find the value of n ._____.
27. In a new system of units, the unit of mass is 100 g, unit of length is 4 m and unit of time is 2 s. Find the numerical value of 10 J in this system. _____
28. If the speed of light (c), acceleration due to gravity (g) and pressure (p) taken as base quantities, then find the magnitude of dimension of g in the dimensional formula of universal gravitational constant.
29. The power for a hovering helicopter depends on its linear size, the density of air and weight per unit volume of the helicopter as $\text{power} \propto (\text{linear size})^x (\text{density of air})^y (\text{weight per unit volume of helicopter})^z$. What is the value of $x + 2y + z$?
30. $\int \frac{dx}{\left(\frac{1}{a}\right)^2 + x^2} = a^n \tan^{-1}(ax)$. Find the value of n .

CHEMISTRY

Max.Marks:100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. Number of atoms in 560 g of Fe (atomic mass = 56g mol^{-1}) is
 A) twice that 70g N B) half that of 20g H C) Both (A) and (B) D) None of these
32. Mass of calcium that has same number of atoms as in 16g of CH_4 is
 A) 200 g B) 100g C) 40g D) 20g
33. For $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, which is the correct mole relationship?
 A) $9 \times \text{mole of Cu} = \text{mole of O}$ B) $5 \times \text{mole of Cu} = \text{mole of O}$
 C) $9 \times \text{mole of Cu} = \text{mole of O}_2$ D) $\text{Mole of Cu} = 9 \times \text{mole of O}$

Space for Rough Work

34. A chunk of sodium metal, Na, if thrown into a bucket of water, produces a dangerously violent explosion from the reaction of sodium with water. If 50.4g of sodium is used, how many moles of sodium will it represent? (Na = 23amu)
- A) 5.22 B) 4.5 C) 2.19 D) 11.8
35. Molality of a solution is the number of
- A) Moles of the solute per 1000g of the solvent
B) Gram equivalent of the solute per kilogram of the solvent
C) Gram moles of the solute per 1000cm^3 of solution
D) Moles of the solute per 100g of the solvent
36. Mol. fraction of the ethanol and water mixture is 0.25. Hence percentage concentration of ethanol by weight of mixture is
- A) 25% B) 75% C) 46% D) 54%
37. The number of significant figures in 0.0045 are
- A) Two B) Three C) Four D) Five.
38. The law of multiple proportions was observed for the pair
- A) CO_2, NO_2 B) $\text{N}_2\text{O}, \text{NO}_2$ C) $\text{H}_2\text{O}, \text{H}_2\text{S}$ D) $\text{H}_2\text{S}, \text{SO}_2$
39. Boron has two isotopes ^{10}B and ^{11}B whose relative abundances are 20% and 80% respectively. Atomic weight of Boron is
- A) 10 B) 11 C) 10.5 D) 10.8
40. If the weight of 5.6 litres of a gas at S.T.P is 11 grams, the gas is (Take Volume = 22.4 L at STP)
- A) Phosphine B) Phosgene C) Nitric oxide D) Nitrous oxide
41. Hydrogen and oxygen combine to form H_2O_2 and H_2O containing 5.93% and 11.2% hydrogen respectively. The data illustrates.
- A) law of conservation of mass B) law of constant proportion
C) law of reciprocal proportion D) law of multiple proportion
42. Total mass before and after the chemical reaction is same. This illustrates the law of
- A) conservation of mass B) constant proportions
C) multiple proportions D) reciprocal proportions

Space for Rough Work

43. What volume of hydrogen gas at 273K and 1 atm pressure will be consumed in obtaining 21.6 g of elemental boron (at. mass = 10.8) from the reduction of boron trichloride with hydrogen:
 A) 44.8 L B) 22.4 L C) 89.6 L D) 67.2 L
44. The equation: $2\text{Al}_{(s)} + \frac{3}{2}\text{O}_{2(g)} \rightarrow \text{Al}_2\text{O}_{3(s)}$ shows that:
 A) 2 mol of Al reacts with $\frac{3}{2}$ mol of O_2 to produce $\frac{7}{2}$ mol of Al_2O_3
 B) 2 g of Al reacts with $\frac{3}{2}$ mol of O_2 to produce one mol of Al_2O_3
 C) 2 g of Al reacts with $\frac{3}{2}$ L of O_2 to produce 1 mol of Al_2O_3
 D) 2 mol of Al reacts with $\frac{3}{2}$ mol of O_2 to produce 1 mol of Al_2O_3
45. If 8 mL of uncombined O_2 remain after exploding O_2 with 4 mL of hydrogen, the number of mL of O_2 originally were-
 A) 12 B) 2 C) 10 D) 4
46. The atomic weight of Cu is 63.546. There are only two naturally occurring isotopes of copper ^{63}Cu and ^{65}Cu . The natural abundance of the ^{63}Cu isotope must be approximately:
 A) 10% B) 30% C) 50% D) 75%
47. A 15 mL sample of 0.20 M MgCl_2 is added to 45 mL of 0.40 M AlCl_3 , what is the molarity of Cl^- ions in the final solution:
 A) 1.0 M B) 0.60 M C) 0.35 M D) 0.30 M
48. A molecule contains 7 carbon atoms 2 oxygen atoms and 9.96×10^{-24} g of other elements. The molar mass of the compound ($N_A = 6.02 \times 10^{23}$)
 A) 122 B) 116 C) 148 D) 154
49. Carbon disulphide ' CS_2 ', can be made from by product SO_2 . The overall reaction is

$$5\text{C} + 2\text{SO}_2 \longrightarrow \text{CS}_2 + 4\text{CO}$$

 How much CS_2 can be produced from 450 kg of waste SO_2 with excess of coke if the SO_2 conversion is 82%.
 A) 219.09 kg B) 120 kg C) 320 kg D) 322.10 kg
50. N_2H_4 , Hydrazine a rocket fuel can be produced according to the following reaction

$$\text{ClNH}_2 + 2\text{NH}_3 \longrightarrow \text{N}_2\text{H}_4 + \text{NH}_4\text{Cl}$$

 When 1000 g ClNH_2 is reacted with excess of NH_3 , 473 g N_2H_4 is produced. What is the % yield of the reaction.
 A) 76.12 % B) 75.10% C) 80% D) 90%

Space for Rough Work

SECTION-II (Numerical Value Answer Type)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. Total number of species present in 1 mole of potash alum in terms of avagadro number, 'N' are
(Given: K = 39u, Al = 27u, S = 32u)
52. The oxide of a metal contains 40% oxygen. The valency of metal is 2. What is the atomic weight of the metal?
53. The volume of water to be added to 400 ml of N/8 HCl to make it exactly N/12 is
54. The normality of orthophosphoric acid having purity of 70% by weight (specific gravity is 1.54) would be
55. Compute the volume of conc. HCl (in ml) (density 1.19g/cm³, 38% HCl by weight) required to make up 18 L of M/50 acid.
56. If the components of air are N₂ 78%, O₂ 21%, Ar 0.9% and CO₂ 0.1% by volume, what would be the molar mass of air? [At. wt of Ar = 40]
57. What is the molarity of a solution in which 160g NaOH is present in 500ml of solution.
58. 20 grams of sulphur on burning in air produced 11.2 L of SO₂ at STP. The percentage of unreacted sulphur_____
59. How many grams H₂SO₄ can be obtained from 1320 gm PbS as per reaction sequence?

$$2\text{PbS} + 3\text{O}_2 \longrightarrow 2\text{PbO} + 2\text{SO}_2$$

$$3\text{SO}_2 + 2\text{HNO}_3 + 2\text{H}_2\text{O} \longrightarrow 3\text{H}_2\text{SO}_4 + 2\text{NO}$$
[At. mass : Pb = 208, S = 32]
60. The density of liquid mercury is 13.6g / cm³. How many moles of mercury are there in 10 litre of the metal? (Atomic mass of Hg = 200)

Space for Rough Work

MATHEMATICS**Max.Marks:100****SECTION - I**
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. The value of $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$ is -

- A) $\frac{3}{8}$ B) $\frac{1}{8}$ C) $\frac{3}{16}$ D) None of these

62. H_1, H_2 are 2 H.M.'s between a, b then $\frac{H_1 + H_2}{H_1 \cdot H_2} =$

- A) $\frac{ab}{a+b}$ B) $\frac{a+b}{ab}$ C) $\frac{a-b}{ab}$ D) $\frac{ab}{a-b}$

63. If $\cot \alpha = 1$ and $\sec \beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha + \beta)$ and the quadrant in which $\alpha + \beta$ lies, respectively are

- A) $-\frac{1}{7}$ and IVth quadrant B) 7 and Ist quadrant C) -7 and IVth quadrant D) $\frac{1}{7}$ and Ist quadrant

64. The value of $2\sin(12^\circ) - \sin(72^\circ)$ is:

- A) $\frac{\sqrt{5}(1-\sqrt{3})}{4}$ B) $\frac{1-\sqrt{5}}{8}$ C) $\frac{\sqrt{3}(1-\sqrt{5})}{2}$ D) $\frac{\sqrt{3}(1-\sqrt{5})}{4}$

65. If $\sin A = n \sin B$, then $\frac{n-1}{n+1} \tan \frac{A+B}{2} =$

- A) $\sin \frac{A-B}{2}$ B) $\tan \frac{A-B}{2}$ C) $\cot \frac{A-B}{2}$ D) None of these

66. If $\sin \theta + \operatorname{cosec} \theta = 2$, then the value of $\sin^{10} \theta + \operatorname{cosec}^{10} \theta$ is

- A) 2 B) 2^{10} C) 2^9 D) 10

67. If $\alpha + \beta + \gamma = 2\theta$, then $\cos \theta + \cos(\theta - \alpha) + \cos(\theta - \beta) + \cos(\theta - \gamma)$ is equal to

- A) $4\sin \frac{\alpha}{2} \cdot \cos \frac{\beta}{2} \cdot \sin \frac{\gamma}{2}$ B) $4\cos \frac{\alpha}{2} \cdot \cos \frac{\beta}{2} \cdot \cos \frac{\gamma}{2}$ C) $4\sin \frac{\alpha}{2} \cdot \sin \frac{\beta}{2} \cdot \sin \frac{\gamma}{2}$ D) $4\sin \alpha \cdot \sin \beta \cdot \sin \gamma$

68. If $\cos(A - B) = \frac{3}{5}$ and $\tan A \tan B = 2$, then which one of the following is true?

- A) $\sin(A + B) = \frac{1}{5}$ B) $\sin(A + B) = -\frac{1}{5}$ C) $\cos(A - B) = \frac{1}{5}$ D) $\cos(A + B) = -\frac{1}{5}$

69. The value of $\cos^2\left(\frac{\pi}{4} + \theta\right) - \sin^2\left(\frac{\pi}{4} - \theta\right)$ is

- A) 0 B) $\cos 2\theta$ C) $\sin 2\theta$ D) $\cos \theta$

Space for Rough Work

70. If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then $\frac{\tan x}{\tan y}$ is equal to
 A) $\frac{a^2}{b^2}$ B) $\frac{a}{b}$ C) $\frac{b}{a}$ D) $\frac{a^2+b^2}{a^2-b^2}$
71. If $a_1, a_2, a_3, \dots, a_n$ be an AP of non-zero terms, then $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{n-1} a_n}$ is equal to
 A) $\frac{n-1}{a_1 a_n}$ B) $\frac{n}{a_1 a_n}$ C) $\frac{n+1}{a_1 a_n}$ D) None of these
72. If sum of an infinite geometric series is $\frac{4}{3}$ and its 1st term is $\frac{3}{4}$, then its common ratio is
 A) $\frac{7}{16}$ B) $\frac{9}{16}$ C) $\frac{1}{9}$ D) $\frac{7}{9}$
73. If $a, a_1, a_2, \dots, a_{2n}, b$ are in arithmetic progression and $a, g_1, g_2, \dots, g_{2n}, b$ are in geometric progression and h is the harmonic mean of a and b , then $\frac{a_1+a_{2n}}{g_1 g_{2n}} + \frac{a_2+a_{2n-1}}{g_2 g_{2n-1}} + \dots + \frac{a_n+a_{n+1}}{g_n g_{n+1}}$ is equal to
 A) $2nh$ B) $\frac{n}{h}$ C) nh D) $\frac{2n}{h}$
74. If $\log_3 2, \log_3 (2^x - 5)$ and $\log_3 (2^x - [7/2])$ are in A.P., then x is equal to
 A) $1/2$ B) 2 C) 3 D) 1
75. Evaluate: $\sum_{k=1}^{11} (2+3^k)$
 A) $\left(22 + \frac{3}{2}(3^{11}-1)\right)$ B) $\left(20 + \frac{3}{2}(3^{11}-1)\right)$ C) $\left(21 + \frac{3}{2}(3^{11}-1)\right)$ D) $\left(22 + \frac{3}{2}(2^{11}-1)\right)$
76. If the $p^{\text{th}}, q^{\text{th}}$ and r^{th} term of an arithmetic sequence are a, b and c respectively, then the value of $[a(q-r) + b(r-p) + c(p-q)] =$ _____
 A) 0 B) pqr C) 1 D) $a+b+c$
77. If $x, 1, z$ are in A.P. and $x, 2, z$ are in G.P., then $x, 4, z$ will be in _____.
 A) AP B) GP C) HP D) None of these
78. Find sum of infinite series $S = 1 + 3r + 5r^2 + 7r^3 + \dots \infty$.
 A) $\frac{1}{(1-r)^2}$ B) $\frac{1-2r}{(1-r)^2}$ C) $\frac{1+r}{(1-r)^2}$ D) $\frac{1+r^2}{(1-r)^2}$
79. The 5th and 11th term of HP are $\frac{1}{10}$ and $\frac{1}{22}$. Find 15th term.
 A) $\frac{1}{30}$ B) $\frac{1}{28}$ C) $\frac{1}{32}$ D) $\frac{1}{26}$

Space for Rough Work

80. Sum of n terms of series $\frac{7}{3^2 \times 4^2} + \frac{9}{4^2 \times 5^2} + \frac{11}{5^2 \times 6^2} + \dots$

A) $\frac{1}{2^2} - \frac{1}{(n+2)^2}$

B) $\frac{1}{3^2} - \frac{1}{(n+3)^2}$

C) $1 - \frac{1}{(n+1)^2}$

D) None of these

SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

81. Let T_r be the r^{th} term of an A.P., for $r = 1, 2, 3, \dots$ if for some positive integers m, n we have $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then T_{mn} equals
82. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x) =$
83. The expression $3 \left[\sin^4 \left(\frac{3\pi}{2} - \alpha \right) + \sin^4 (3\pi + \alpha) \right] - 2 \left[\sin^6 \left(\frac{\pi}{2} + \alpha \right) + \sin^6 (5\pi - \alpha) \right]$ is equal to
84. The expression $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to
85. If $y = (1 + \tan A)(1 - \tan B)$, where $A - B = \frac{\pi}{4}$, then $(y+1)^{y+1}$ is equal to
86. The value of $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 360^\circ$ is equal to
87. If $\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \dots + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$, then the remainder when K is divided by 6 is
88. Let $A_1, A_2, A_3, \dots$ be an increasing geometric progression of positive real numbers. If $A_1 A_3 A_5 A_7 = \frac{1}{1296}$ and $A_2 + A_4 = \frac{7}{36}$, then the value of $A_6 + A_8 + A_{10}$ is equal to
89. If AM and GM between two numbers are 16 and 8 respectively, then HM of number is
90. The number of numbers that are divisible by 9 between 1 and 1000 is _____.

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Space for Rough Work